xVA

1. **On Day 2 EOD, the MTM becomes V2 = −200,000. Which party posts VM at Day 2 EOD, and what is the VM amount ΔC2?**

   C2 = V2 = −200,000
   ΔC2 = C2 − C1 = −200,000−0 = −200,000
   A negative collateral balance means Bank A must post VM amount of $200,000

2. **Alternatively, suppose on Day 2 EOD the MTM is V2 = +150,000. Assume Bank A can invest any received cash at its unsecured funding/investment rate rF = 6%. Compute the funding benefit over Day 2 (one day of interest).**

   C2 = V2 = 150,000
   ΔC2 = C2 − C1 = 150,000
   So, **Client B posts $150,000 to Bank A**

   Bank A receives 150k and pays collateral interest at 4% and invests cash at 6%.
   Net funding benefit over 1 Day:
   Benefit=150,000×(rF−rC)x (1/360)
   = 150,000×(0.06−0.04)×(1/360) = $ 8.33
   Thus, fund benefit is $8.33 for Day 2.

Exposure Calculation for a Bond Forward

Q1. In assignment 1, you have worked out the carry method to price a bond forward contract, i.e., basically,

- **Repo the spot bond (t=0) to settlement date (T) at rR;**
- **Subtract the coupon reinvested at repo rate rR paid between 0 and T;**
- **discount the payoff at un-secured funding rate rF1;**

For exposure, we need to perform the above steps at time t. This requires simulation some relevant dynamics. Claim: we need at least three stochastic factors X1, X2, and XF that enables separating bond yield, repo rate and unsecured funding rate. Give your justification of this claim

We need at least three stochastic factors because the forward value depends on three economically distinct risk drivers. These drivers move independently of each other.

1. Bond yield dynamics determine the spot bond price and MTM volatility.
2. Repo rate dynamics determine carry and reinvestment value of coupons.
3. Unsecured funding rate dynamics determine discounting and funding valuation adjustment.

The three quantities $B(t;t,T_M)$, $A(t,T_S)$and $P_F(t,T_S)$ appearing in the carry formula are driven by **three economically distinct rate processes**. No two of them can be collapsed into a single factor without destroying the spread information that xVA fundamentally depends on. Hence **at least three stochastic factors** X_1, X_2, $X_F$are required.

Q2. Let us adopt a multi-factor gaussian model, in which case, a i-discount factor has a closed form in terms of the state variables, Pi(t,T) = exp Ai(t,T)−Bi(t,T)⊤X(t) (2) and X(t) follows some known dynamics (Ornstein–Uhlenbeck process), thus Ai and Bi are all known. Is express V (t) in terms of Pi(t, T) known at time t or time T ? Can you Pi(t, ·)'s ?

- Discount factor $P^i(t,T) = \exp\left\{A^i(t,T) - B^i(t,T)^T X(t)\right\}$
- $X(t)$ follows an OU process
  A^i, B^i are deterministic functions
  $X(t)$ is fully known at t

  Therefore $P_i(t,T)$ is fully known at time t

$V(t)$ in terms of $P^i(t,\cdot)$

1. Spot Bond Price
   $$B(t) = \sum_{T_i > t} c_i P^1(t,T_i) + 100 P^1(t,T_m)$$
   $P^1(t,T_i) \leftarrow$ is discount factor corresponding to Bond Yield

2. Repo Accrual factor
   The reinvesting of c_i from T_i to T_S is governed by repo curve.
   $$A(T_i,T_S) = \frac{P^2(t,T_i)}{P^2(t,T_S)} \quad (P^2 - \text{repo curve})$$

3. Bond Forward Price
   $$B(t,T_S,T_m) = \frac{B(t)}{P^2(t,T_S)} - \sum c_i \frac{P^2(t,T_i)}{P^2(t,T_S)}$$

4. Since $B(t;T_S,T_m)$ is F_t measurable, the MTM value can obtained by discounting by P^F (unsecured funding rate)
   $$V(t) = \frac{N}{100} \times P^F(t,T_S) \times \left[\frac{\sum_{T_i > t} c_i P^1(t,T_i) + 100 P^1(t,T_m)}{P^2(t,T_S)} - \sum_{T_i < T_S} c_i \frac{P^2(t,T_i)}{P^2(t,T_S)} - K \right]$$

Q3. Following the notes, finish the steps to compute CVA for a long position with this contract. (This is just to make sure you understand the procedures to calculate xVA based given exposure profile)

For the long position and assuming zero collateralization the exposure is

$$E(t) = \max\{V(t), 0\}$$

and corresponding expected exposure is at time ti is,

$$EE(ti) = E^{Q}[\max\{V(ti), 0\}]$$

model the counterparty default time by using the hazard rate h(t)

$$S(t) = \exp(-\int_0^t h(s))$$

where S(t) is survival probability

and the probability of default over (t_{i-1},t_i)

$$\Delta PD(t_i) = S(t_{i-1}) - S(t_i)$$

Finally with recovery rate R the CVA calculation is,

$$\mathbf{CVA = (1\text{-}R)\sum_{i=1}^{N} P(0, ti) EE(ti)\, \Delta PD(ti)}$$

Where discounting factor $P_i(t, T) = \exp\{A^i(t,T) - B^i(t,T)^\top X(t)\}$

Q4. As a follow up question: how would you come up with a sensible hazard rate that is factored into your survival probability calculation ? (Suppose you're buying from a bank with certain credit rating)

A sensible hazard rate should be derived from market-implied credit risk, using the bank's CDS term structure.

$$h = \frac{CDS\ spread}{1 - R}$$

If CDS is unavailable, bond spreads or rating-based historical default frequencies can be used as alternatives. The resulting hazard rate produces a survival probability curve that is risk-neutral, forward-looking, and consistent with market pricing, which is required for CVA.